LABORATORY TITLE: ANDROID FORENSICS INVESTIGATION OF A GOOGLE PIXEL 3

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COURSE NAME: MOBILE AND IOT FORENSICS CASE STUDY

COURSE CODE: SBT-DF205 (LAB 1)

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VERSION: Version 1.0

Executive Summary

The supplied extraction was examined through system configuration files, Android user and account records, SQLite databases, application preferences, communication records, Google Maps databases and image metadata. The examination identified two Android user profiles, account associations across several applications, SMS and MMS records, call-log entries, Google Messages and Dialer records, WhatsApp and TextNow artefacts, saved Google Maps place records, and photographs containing capture-time and GPS metadata. Cross-source comparisons provided corroboration between Google Messages and Android SMS records, between Google Dialer and Android call records, and between two Google Maps databases. A sequence of call, SMS and MMS records was also identified on 14 February 2020.

However, the findings do not independently establish that the communications or information sharing were unauthorized. Account associations and contact names do not conclusively identify the person who performed an action, and saved location records do not prove physical presence. The available evidence therefore supports an unresolved conclusion regarding the allegation. This report is limited to the supplied extraction and the artefacts examined. The original evidence archive's hash and a complete formal chain of custody were not verified in the recorded examination.

Forensic Methodology and Tools

The examination used file-system artefacts from the supplied Pixel 3 extraction. Configuration files were inspected for device and profile information. SQLite databases were examined for table structures, record counts and relevant entries. Application preferences were reviewed for version and account-association information. Photo metadata was examined using ExifTool, and SHA-256 hashes were recorded for selected files.

The examination environment and tool versions recorded were:

Python – 3.14.7

SQLite - 3.46.1

ExifTool – 13.55

GNU sha256su - 9.10

7-Zip - 26.02

The recorded working directory was /home/kali/Digital-Forensics-ICDFA/pixel3. The examination date and time recorded by the system command was 29 October 2026, 04:11:34 UTC.

The examination was based on the extracted files available in the working directory. The findings therefore describe records present in the supplied extraction, not necessarily every activity that occurred on the physical device.

Findings

Task 1 - Evidence Preservation and Integrity

The examination was conducted using the extracted Pixel 3 data in the working directory. SHA-256 hashes were calculated for selected databases and image files to provide integrity references for those files.

The following hashes were recorded:

File	SHA-256
WifiConfigStore.xml under data/misc/wifi/	cbd2b0039aa786967740b182c4229bf9e0ce147640f0bc00b93255a9c8613986
WifiConfigStore.xml under	3737a8cfbb5a8516ef7112673662e428cf37edf71723deef87b3cc5167bc5b8c
Primary mmssms.db	cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342
User-de copy of mmssms.db	d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827
IMG_20200213_195030.jpg	2ed43a0a7b01a58f89c23263542282d3d5713b5cd2f22d7b3d2f56a109145ea6
IMG_20200209_132353.jpg	308df52bdab5aa52e9e82b2a853049c688450c2ff6ad76cd75f0c063e61853fa
IMG_20200201_142207.jpg	e8bc8fd6a1a8843961e7416b65bd203ca9c6b8a1a1f6375c2b0a15b552dc9100
gmm_myplaces.db	9d09ce47a0c03bd5ffaad8a3869e0852d430e915dad852d08855b7a8ca47f7d8
gmm_sync.db	2e96f7a1ab550b26b13675ca972f559540c955f52b6e55dd4c71de39d2f59ce2

These hashes identify the examined versions of the listed files.

Task 2 - Device Identity and User Profiles

The system builds properties identify the extracted system as a Google/Pixel 3 (blueline) Android 10 build. Relevant properties included:

- Android release: 10
- SDK version: 29
- Build ID: QQ1A.200105.003
- Build incremental: 6079943
- Security patch level: 2020-01-01
- Build type: user
- Product: blueline

The build properties support identifying the extraction as belonging to a Pixel 3 Android 10 environment. Some individual properties, such as the generic device and mainline model strings, are not sufficient on their own to identify the commercial device model.

The Android user records identified two profiles:

- User ID 0, with the profile name This Is.
- User ID 11, with the profile name User 2.

Both profiles contained the same Android build fingerprint. Account databases associated User 0 with several applications and services, including Google, WhatsApp and TextNow, alongside other messaging applications. User 11 had a separate Google account association.

The profile records include last-login timestamps. These indicate recorded profile activity, but they do not independently identify the person who operated the device. Mirrored copies of user-profile files were not counted as additional users.

The extraction contains two Android user profiles and multiple application/account associations. These records provide context for interpreting the artefacts but do not conclusively establish the real-world identity of the device operator.

Task 3 - System Settings and Connectivity

Android settings files and Wi-Fi and Bluetooth artefacts were examined.

Relevant values in the User 0 global settings included:

- wifi_on=1
- bluetooth_on=0
- airplane_mode_on=0
- mobile_data=1
- development_settings_enabled=1
- adb_enabled=1
- auto_time=1
- auto_time_zone=1
- Device name: Pixel 3

The secure settings included location_mode=3, with gps,network listed as location providers. Other values included backup_enabled=0 and mock_location=0.

The Wi-Fi configuration files contained configuration-store structures, but the available examination did not establish a complete history of network connections. The Bluetooth database contained metadata-related tables, but no dedicated connection-history table was identified in the examined schema.

The settings values represent stored configuration at the time represented by the extraction. They do not prove that a feature was actively used during a particular communication event. Likewise, the presence of developer options or ADB settings is not, on its own, proof of unauthorized access.

The artefacts show stored system configuration, including Wi-Fi, mobile data, location and development settings. They do not establish a complete connectivity history or demonstrate that these settings were used to facilitate the suspected incident.

Task 4 - Android Communications, Contacts and Calendar

The primary Android telephony database, mmssms.db, contained:

Table	Records
sms	24
pdu	7
part	10
threads	17

The primary database was 118,784 bytes. A copy in the user-de directory was also examined and had no records in the corresponding SMS, PDU, part and thread tables.

The SMS records included a message dated 9 February 2020 at 19:01:42.392 UTC: “I am on my way to the coffee shop. Do you want anything?” A corresponding MMS record dated 19:03:08 UTC contained “No, thank you”. Another SMS dated 14 February 2020 at 13:09:03.277 UTC stated: “Wow, the messages sync over between accounts.” An MMS record at 13:10:08 UTC contained “That is interesting.”

The database also contained call records spanning 31 January to 14 February 2020. Some records contained phone numbers without contact names, while later records were associated with the saved contact Josh Hickman. Different number formats appeared in the communications and contact records.

The contacts database contained four raw contact rows grouped under one contact ID for Josh Hickman. This is consistent with multiple underlying contact representations or sources and should not be interpreted as four distinct people.

The examined calendar database contained one calendar but no event records in the principal event-related tables inspected. This means no calendar events were found in those tables; it does not establish that the device had never been used for scheduling.

Android communications records establish that SMS, MMS and call-log artefacts were present in the extraction. Contact associations assist with interpretation but do not prove the identity of the person participating in a communication or the purpose of the communication.

Task 5 - Google Messages and Google Dialer

The Google Messages database, bugle_db, contained 36 records in the messages table. It also contained conversation, participant, part and other application tables. The database had associated WAL and SHM files, which were recognized as relevant SQLite sidecars.

Of the Google Messages records, 24 were linked to AOSP SMS records through the sms_message_uri field. Three non-sensitive message comparisons agreed with the corresponding AOSP records. Duplicate representations of the same message were treated as corroboration rather than counted as separate communications. Other message comparisons were not fully reported because the relevant content was withheld from the comparison.

The Google Dialer annotated_call_log.db contained 12 records. Twelve records matched AOSP call-log records by timestamp, duration and call type, with identical timestamps in the comparisons recorded. Five matched records agreed in their primary number fields. Other records showed differences in number fields while raw or formatted number fields agreed. Two AOSP call records did not have matching records in the examined Dialer table.

The separate dialer.db contained smart-dial and other application data but no call-history table was identified in the examined schema.

Google Messages and Google Dialer provide useful corroboration for records in the Android communication databases. Differences and unmatched records remain unexplained by the available evidence and should not be assigned a cause without further examination.

Task 6 - WhatsApp and TextNow

WhatsApp

The WhatsApp extraction included msgstore.db, wa.db, associated SQLite sidecars and application preference files. The msgstore.db contained two call-log records, three chat records and 19 records in the legacy messages table. Other message, media, thumbnail and search-related tables were also present. The wa.db contained contact and business-profile records.

The application preferences indicated version 2.20.11 and the stored profile name ThisIsDFIR. The examined records included messages from late January and early February 2020. Some records contained text, while others had no text content in the fields examined. One message record referenced an image filename.

The WhatsApp contact database associated a stored display name, Josh Hickman, with one messaging identifier. Such an association is useful for analysis but does not independently prove who controlled that account or authored each message.

TextNow

The TextNow database contained 15 contact records, one conversation and nine message records. The application preferences contained account-related information, including an enabled account status and stored username. Sensitive account identifiers and message content have been omitted from this report.

The examined messages included exchanges dated 8 February 2020, including “Hi there!”, “Back at it again, I see.” and “Yep! Have to keep generating data.” Other records included system or event-type entries and messages for which the content was not reproduced.

The direction and type fields were recorded in the database, but they have not been interpreted as incoming or outgoing in this report without a confirmed mapping for the relevant schema and application version.

WhatsApp and TextNow contain application-level communication and account artefacts. These records establish stored application activity but do not independently establish that the activity was unauthorized or identify the human operator.

Task 7 - Google Maps and Photo Metadata

Several Google Maps databases were identified, including gmm_myplaces.db, gmm_sync.db, gmm_storage.db and ugc_photos_location_data.db.

The gmm_myplaces.db database contained a saved-place record for 121 E Tryon Rd, Raleigh, North Carolina 27603, with stored coordinates and a timestamp corresponding to 4 April 2019 at 19:49:03.604 UTC. The same address, coordinates and timestamp were found in gmm_sync.db, providing cross-database corroboration.

Two other address-bearing records were found in synchronization data: one associated with Holly Springs, North Carolina, and another with Secaucus, New Jersey. Their queried timestamps were zero and therefore did not provide usable event times.

The two tables examined in ugc_photos_location_data.db contained no records. This finding is limited to those tables and does not establish that no location-related information existed anywhere else in the extraction.

Three photographs were examined:

- IMG_20200209_132353.jpg: EXIF capture time of 9 February 2020 at 13:23:53.139167 with a -05:00 offset, equivalent to 18:23:53.139167 UTC. GPS metadata was present.
- IMG_20200213_195030.jpg: EXIF capture time of 13 February 2020 at 19:50:31.590730 with a -05:00 offset, equivalent to 14 February 2020 at 00:50:31.590730 UTC. GPS metadata was present.
- IMG_20200201_142207.jpg: no EXIF capture-time or GPS fields were displayed in the examined output.

The two photographs with GPS metadata contained coordinates, altitude and GPS time information. Their coordinates did not match the saved Raleigh address coordinates. The file modification times were not treated as substitutes for capture times.

The Maps databases contain saved-place and synchronization records, while two photographs contain capture-time and GPS metadata. These artefacts establish stored metadata and place associations but do not independently prove a visit, physical presence, or the identity of the person who created the records.

Task 8 - Timeline and Cross-Source Analysis

The timeline was reconstructed by converting recorded timestamps to UTC using their identified formats. AOSP SMS and call timestamps were treated as milliseconds, MMS PDU timestamps as seconds, and the two photographs' EXIF capture times were converted using the recorded -05:00 offset.

The timeline includes at least 12 distinct candidate events from four artefact families: Android communications, Google Maps, photo metadata and application-level communications. The accompanying Timeline.csv should retain each raw timestamp, field name, unit or format, converted UTC time, description and evidence ID.

Three important cross-source comparisons were established:

1. Google Messages and AOSP SMS: 24 Google Messages records were linked to AOSP SMS records; three non-sensitive comparisons agreed. These are corroborating records, not additional message events.
2. Google Dialer and AOSP calls: 12 annotated-call records matched AOSP call records on timestamp, duration and call type. Two AOSP call records were unmatched in the examined Dialer table.
3. Call, SMS and MMS sequence: a call associated with Josh Hickman at 13:05:57.104 UTC on 14 February 2020 was followed by an SMS at 13:09:03.277 UTC and an MMS at 13:10:08 UTC. The time sequence does not prove that all records were part of the same conversation.
4. Google Maps databases: the Raleigh address record was present in both gmm_myplaces.db and gmm_sync.db with matching coordinates and timestamp. This corroborates the stored place record but not a physical visit.

The timeline contains records that can be ordered chronologically, but the available data does not establish the complete context or author of every event. Duplicate records were not counted as separate events.

Assessment of the Allegation

The allegation concerns suspected unauthorized communication and sharing of information. The examination identified stored communications across Android messaging records and third-party applications, as well as account associations, call records, photo metadata and saved location records.

However, the examination did not identify evidence sufficient to determine that the communications were unauthorized. The presence of messages or multiple messaging applications does not itself establish a policy violation. Likewise, a stored contact name or application account does not prove who performed an action.

Two alternative explanations remain plausible:

1. Routine or authorized activity: the communications may have been ordinary or approved activity. The available extraction does not establish the relevant company authorization rules or the full business context.
2. Account synchronization or shared-profile activity: some records may reflect application synchronization or activity associated with different profiles or accounts. The available records do not conclusively attribute each action to one individual.

Additional evidence that could help resolve the allegation includes the applicable company policy and authorization records, fuller context for relevant messages and media, and independently corroborating organizational or service-side records obtained lawfully.

Conclusion

The examination establishes the presence of communications, application records, account associations, saved place records and photo metadata. Cross-source matches support the consistency of several stored records. However, the available evidence does not independently prove unauthorized communication or information sharing, nor does it conclusively identify the person responsible for each recorded action.

This conclusion does not establish that misconduct occurred, and it does not establish that misconduct did not occur. It reflects the limits of the available evidence.

Limitations

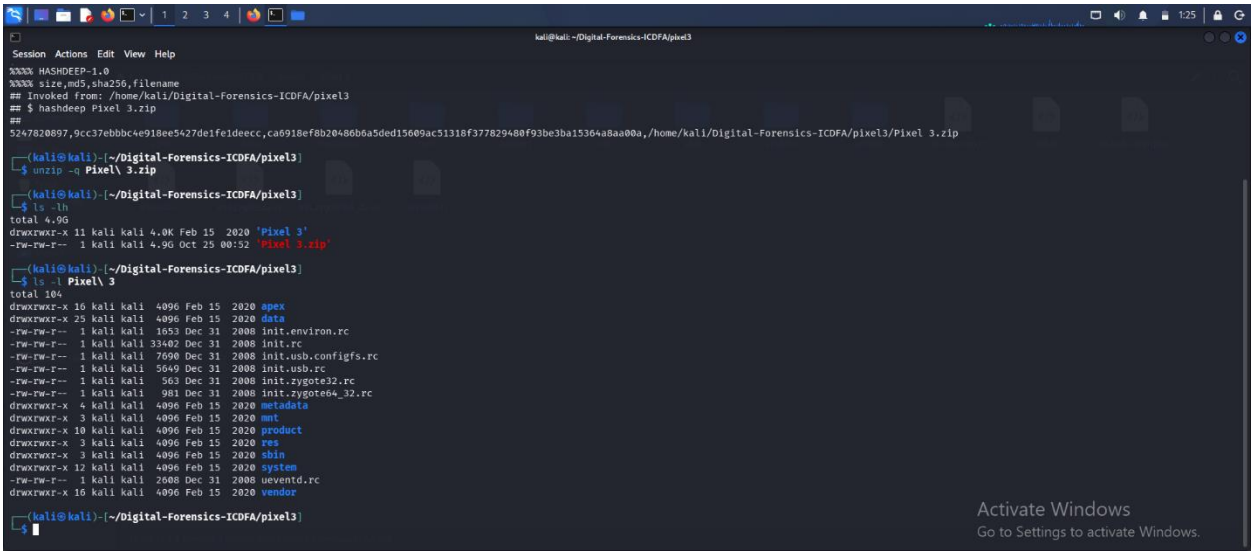
- The examination was based on a supplied extraction rather than a live-device examination.
- The original archive hash and complete formal chain of custody were not verified in the recorded results.
- Hashes were recorded for selected files, not every file in the extraction.
- The available results do not provide complete message content and context for every application record.
- Some call-number fields differed across databases, and two AOSP call records were not matched in the examined Dialer table.
- Application account and contact associations do not prove the identity of the human operator.

- Saved Maps records do not prove physical presence at a location.
- Empty tables show only that no records were found in those specific tables.
- The report's findings are limited to the files, fields and records actually examined.

Screenshots

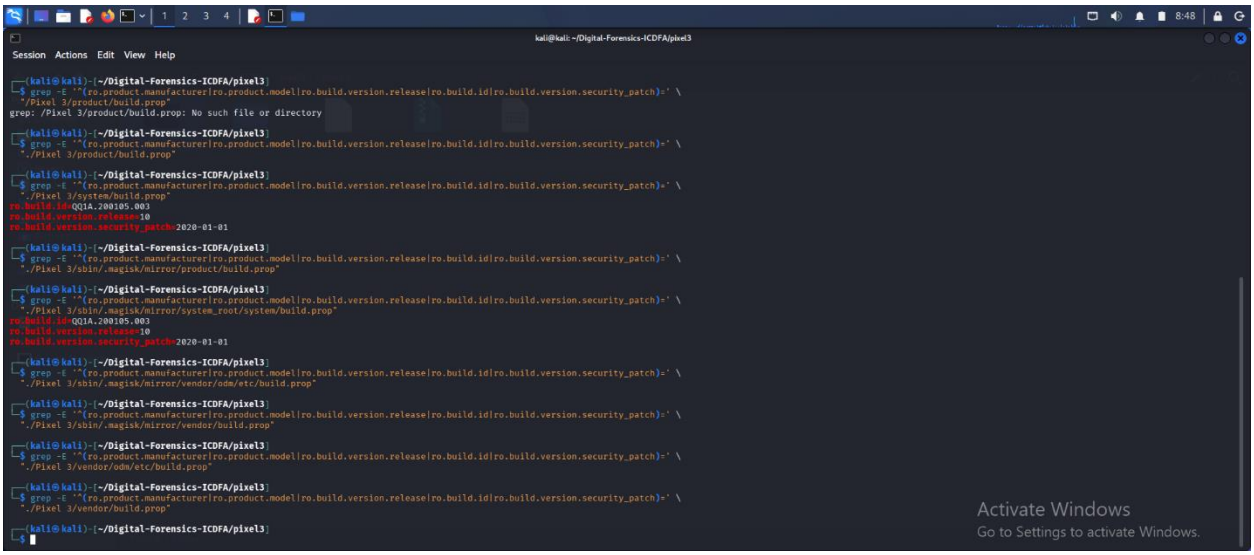
Task 1 Evidence preservation

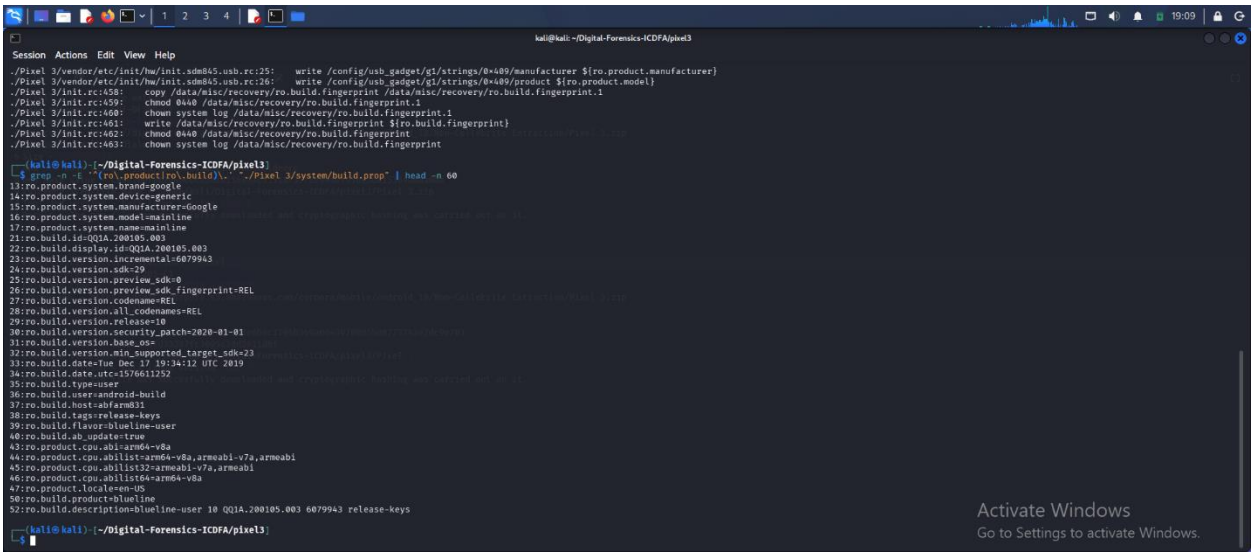


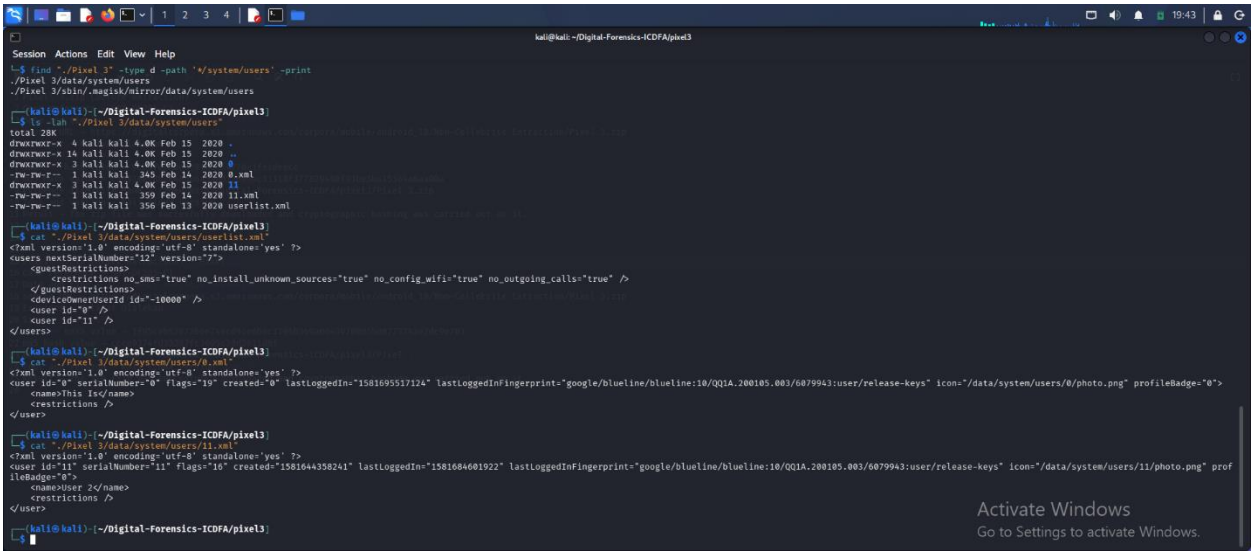


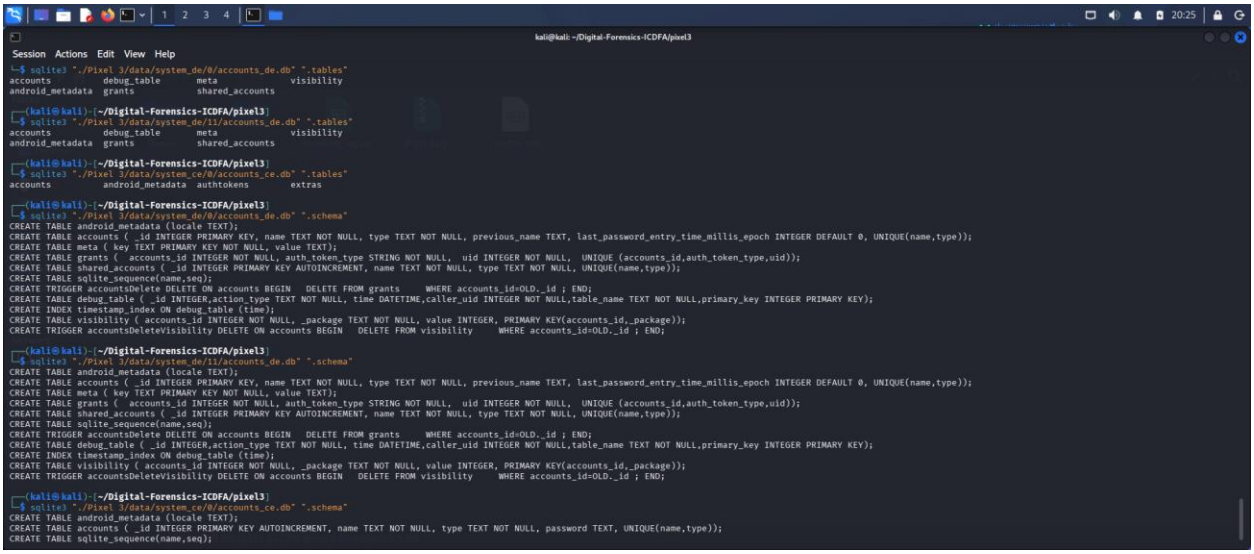
Task 2 Device identity and profiles

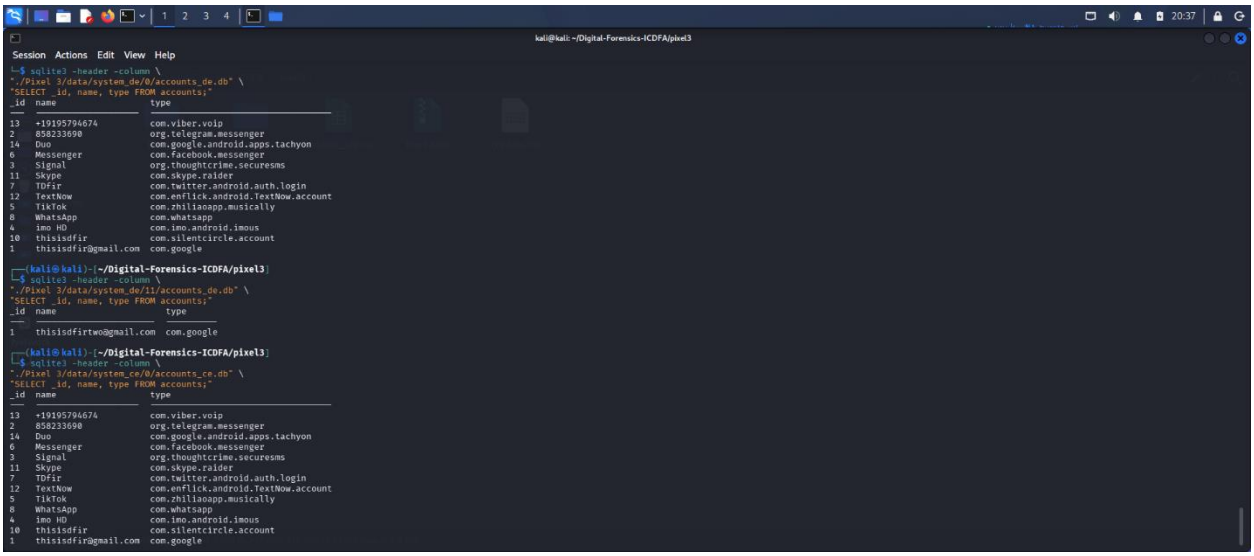



kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"/Pixel 3/product/build.prop"
grep: /Pixel 3/product/build.prop: No such file or directory
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/product/build.prop"
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/system/build.prop"
ro.build.id=QQ1A.200105.003
ro.build.version.release=10
ro.build.version.security_patch=2020-01-01
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/sbin/.magisk/mirror/product/build.prop"
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/sbin/.magisk/mirror/system_root/system/build.prop"
ro.build.id=QQ1A.200105.003
ro.build.version.release=10
ro.build.version.security_patch=2020-01-01
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/sbin/.magisk/mirror/vendor/odm/etc/build.prop"
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/sbin/.magisk/mirror/vendor/build.prop"
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/vendor/odm/etc/build.prop"
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -E '^(ro.product.manufacturer|ro.product.model|ro.build.version.release|ro.build.id|ro.build.version.security_patch)=' \
"./Pixel 3/vendor/build.prop"
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$
Activate Windows
Go to Settings to activate Windows.

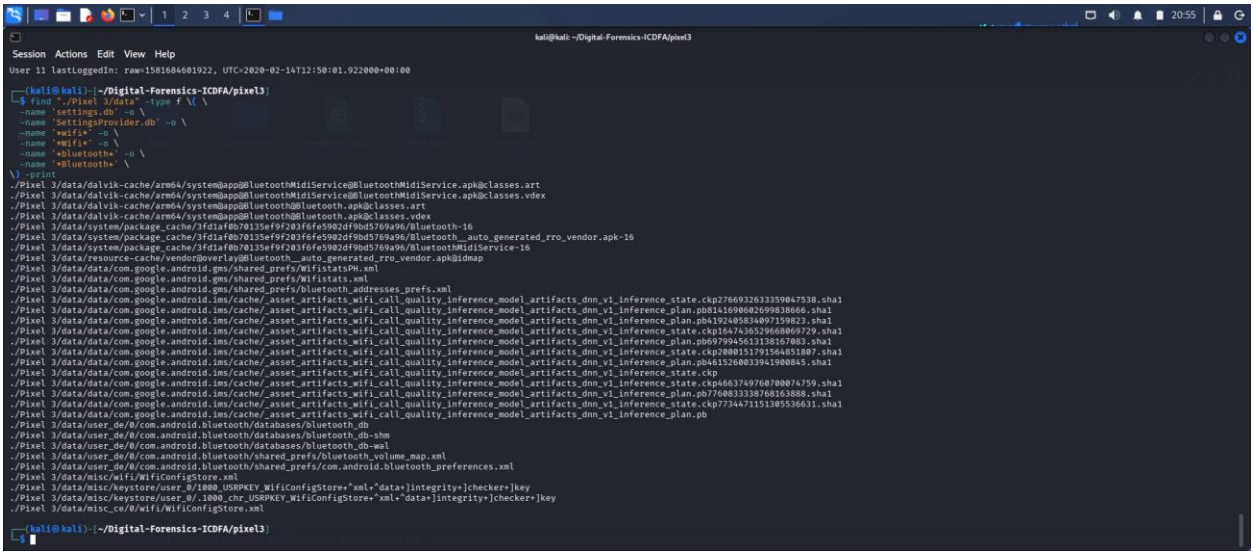

kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
./Pixel 3/vendor/etc/init/hw/init.sdm845.usb.rc:25: write /config/usb_gadget/g1/strings/0x409/manufacturer ${ro.product.manufacturer}
./Pixel 3/vendor/etc/init/hw/init.sdm845.usb.rc:26: write /config/usb_gadget/g1/strings/0x409/product ${ro.product.model}
./Pixel 3/init.rc:458: copy /data/misc/recovery/ro.build.fingerprint /data/misc/recovery/ro.build.fingerprint.1
./Pixel 3/init.rc:459: chmod 0440 /data/misc/recovery/ro.build.fingerprint.1
./Pixel 3/init.rc:460: chown system log /data/misc/recovery/ro.build.fingerprint.1
./Pixel 3/init.rc:461: write /data/misc/recovery/ro.build.fingerprint ${ro.build.fingerprint}
./Pixel 3/init.rc:462: chmod 0440 /data/misc/recovery/ro.build.fingerprint
./Pixel 3/init.rc:463: chown system log /data/misc/recovery/ro.build.fingerprint
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ grep -n -E '^(ro\.product|ro\.build)\.' "./Pixel 3/system/build.prop" | head -n 60
13:ro.product.system.brand=google
14:ro.product.system.device=generic
15:ro.product.system.manufacturer=Google
16:ro.product.system.model=mainline
17:ro.product.system.name=mainline
21:ro.build.id=QQ1A.200105.003
22:ro.build.display.id=QQ1A.200105.003
23:ro.build.version.incremental=6079943
24:ro.build.version.sdk=29
25:ro.build.version.preview_sdk=0
26:ro.build.version.preview_sdk_fingerprint=REL
27:ro.build.version.codename=REL
28:ro.build.version.all_codenames=REL
29:ro.build.version.release=10
30:ro.build.version.security_patch=2020-01-01
31:ro.build.version.base_os=
32:ro.build.version.min_supported_target_sdk=23
33:ro.build.date=Tue Dec 17 19:34:12 UTC 2019
34:ro.build.date.utc=1576611252
35:ro.build.type=user
36:ro.build.user=android-build
37:ro.build.host=abfarm831
38:ro.build.tags=release-keys
39:ro.build.flavor=blueline-user
40:ro.build.ab_update=true
43:ro.product.cpu.abi=arm64-v8a
44:ro.product.cpu.abilist=arm64-v8a,armeabi-v7a,armeabi
45:ro.product.cpu.abilist32=armeabi-v7a,armeabi
46:ro.product.cpu.abilist64=arm64-v8a
47:ro.product.locale=en-US
50:ro.build.product=blueline
52:ro.build.description=blueline-user 10 QQ1A.200105.003 6079943 release-keys
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$
Activate Windows
Go to Settings to activate Windows.


kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
$ find "./Pixel 3" -type d -path '*/system/users' -print
./Pixel 3/data/system/users
./Pixel 3/sbin/.magisk/mirror/data/system/users
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ ls -lah "./Pixel 3/data/system/users"
total 28K
drwxrwxr-x 4 kali kali 4.0K Feb 15 2020 .
drwxrwxr-x 14 kali kali 4.0K Feb 15 2020 ..
drwxrwxr-x 3 kali kali 4.0K Feb 15 2020 0
-rw-rw-r-- 1 kali kali 345 Feb 14 2020 0.xml
drwxrwxr-x 3 kali kali 4.0K Feb 15 2020 11
-rw-rw-r-- 1 kali kali 359 Feb 14 2020 11.xml
-rw-rw-r-- 1 kali kali 356 Feb 13 2020 userlist.xml
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ cat "./Pixel 3/data/system/users/userlist.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
<guestRestrictions>
<restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
</guestRestrictions>
<deviceOwnerUserId id="-10000" />
<user id="0" />
<user id="11" />
</users>
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ cat "./Pixel 3/data/system/users/0.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/system/users/0/photo.png" profileBadge="0">
<name>This Is</name>
<restrictions />
</user>
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ cat "./Pixel 3/data/system/users/11.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/system/users/11/photo.png" profileBadge="0">
<name>User 2</name>
<restrictions />
</user>
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$
Activate Windows
Go to Settings to activate Windows.





Task 3 Settings and connectivity



```
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ sha256sum \
"./Pixel 3/data/misc/wifi/WifiConfigStore.xml" \
"./Pixel 3/data/misc_ce/0/wifi/WifiConfigStore.xml"
cbd2b0039aa78696774b1822c4229bf9e0ce14764ff0bc00b93255a9c8613986  ./Pixel 3/data/misc/wifi/WifiConfigStore.xml
3737a8cfbb5a8516ef7112673662e428cf37edf71723deef87b3cc5167bc5b8c  ./Pixel 3/data/misc_ce/0/wifi/WifiConfigStore.xml

┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ stat -c '%n | Size: %s bytes' \
"./Pixel 3/data/misc/wifi/WifiConfigStore.xml" \
"./Pixel 3/data/misc_ce/0/wifi/WifiConfigStore.xml"
./Pixel 3/data/misc/wifi/WifiConfigStore.xml | Size: 626 bytes
./Pixel 3/data/misc_ce/0/wifi/WifiConfigStore.xml | Size: 966 bytes

┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$
```

```
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ sqlite3 "./Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db" ".tables"
android_metadata   metadata           room_master_table

┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ sqlite3 "./Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db" ".schema"
CREATE TABLE android_metadata (locale TEXT);
CREATE TABLE room_master_table (id INTEGER PRIMARY KEY,identity_hash TEXT);
CREATE TABLE `metadata` (`address` TEXT NOT NULL, `migrated` INTEGER NOT NULL, `a2dpSupportsOptionalCodecs` INTEGER NOT NULL, `a2dpOptionalCodecsEnabled` INTEGER NOT NULL, `a2dp_priority` INTEGER, `a2dp_sink_priority` INTEGER, `hfp_pri
ority` INTEGER, `hfp_client_priority` INTEGER, `hid_host_priority` INTEGER, `pan_priority` INTEGER, `pbap_priority` INTEGER, `pbap_client_priority` INTEGER, `map_priority` INTEGER, `sap_priority` INTEGER, `hearing_aid_priority` INTEGER
, `map_client_priority` INTEGER, `manufacturer_name` BLOB, `model_name` BLOB, `software_version` BLOB, `hardware_version` BLOB, `companion_app` BLOB, `main_icon` BLOB, `is_untethered_headset` BLOB, `untethered_left_icon` BLOB, `untethe
red_right_icon` BLOB, `untethered_case_icon` BLOB, `untethered_left_battery` BLOB, `untethered_right_battery` BLOB, `untethered_case_battery` BLOB, `untethered_left_charging` BLOB, `untethered_right_charging` BLOB, `untethered_case_cha
rging` BLOB, `enhanced_settings_ui_uri` BLOB, PRIMARY KEY(`address`));

┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ sqlite3 -header -column \
"./Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db" \
"SELECT COUNT(*) AS metadata_records FROM metadata;"
metadata_records
────────────────
2

┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ sqlite3 -header -column \
"./Pixel 3/data/user_de/0/com.android.bluetooth/databases/bluetooth_db" \
"SELECT
 COUNT(*) AS total_records,
 SUM(CASE WHEN manufacturer_name IS NOT NULL AND length(manufacturer_name)>0 THEN 1 ELSE 0 END) AS records_with_manufacturer,
 SUM(CASE WHEN model_name IS NOT NULL AND length(model_name)>0 THEN 1 ELSE 0 END) AS records_with_model
 FROM metadata;"
total_records  records_with_manufacturer  records_with_model
─────────────  ─────────────────────────  ─────────────────
2              0                          0

┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$
```

Task 4 AOSP communications and contacts

```
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ find "./Pixel 3/data" -type f \( \
  -name 'mmssms.db' -o \
  -name 'contacts2.db' -o \
  -name 'calllog.db' \
\) -print
./Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db
./Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db
./Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db
./Pixel 3/data/user_de/0/com.android.providers.telephony/databases/mmssms.db

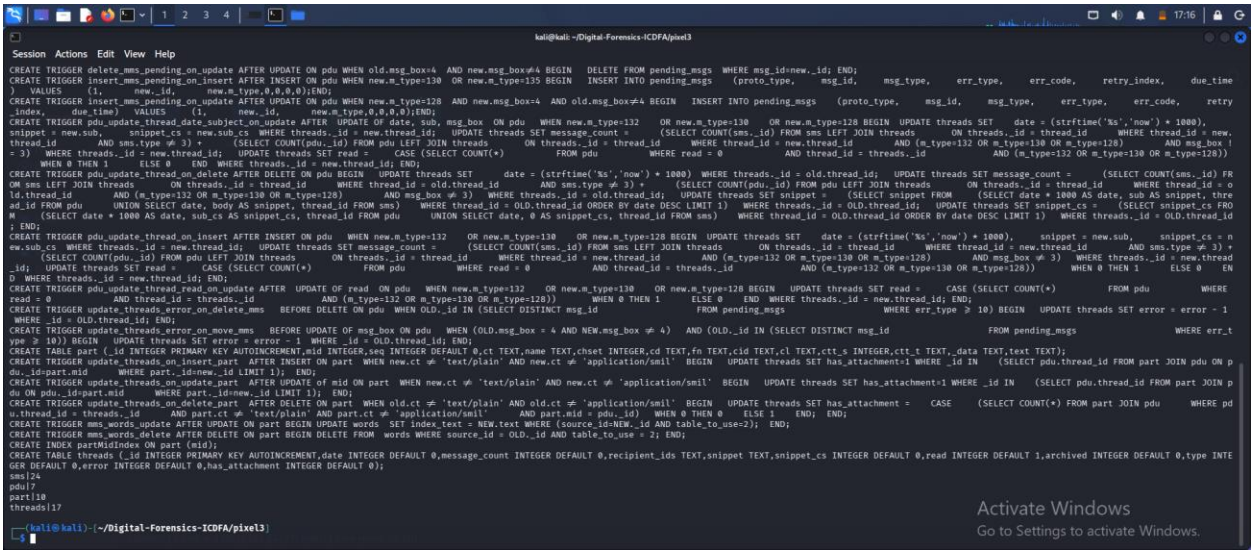
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ find "./Pixel 3/data/data/com.android.providers.telephony/databases" \
    "./Pixel 3/data/user_de/0/com.android.providers.telephony/databases" \
    -maxdepth 1 -type f \
    \( -name 'mmssms.db*' \) \
    -printf '%p | %s bytes\n'
./Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db | 118784 bytes
./Pixel 3/data/user_de/0/com.android.providers.telephony/databases/mmssms.db | 118784 bytes

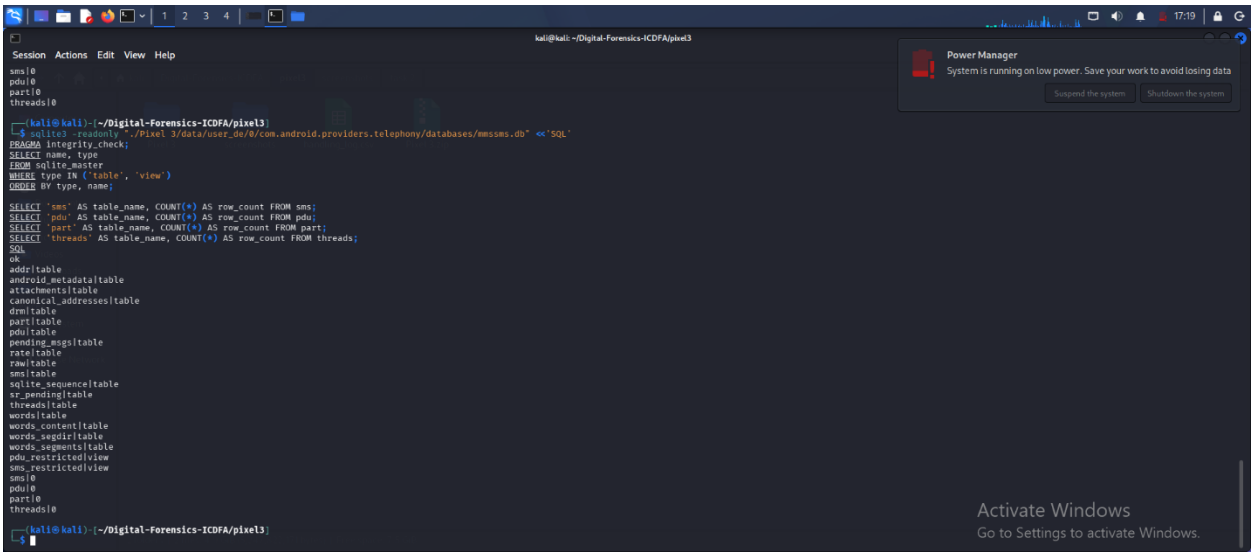
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ find "./Pixel 3/data/data/com.android.providers.telephony/databases" \
    "./Pixel 3/data/user_de/0/com.android.providers.telephony/databases" \
    -maxdepth 1 -type f \
    \( -name 'mmssms.db*' \) \
    -exec sha256sum {} \;
cd5452f529331252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  ./Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db
d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827  ./Pixel 3/data/user_de/0/com.android.providers.telephony/databases/mmssms.db

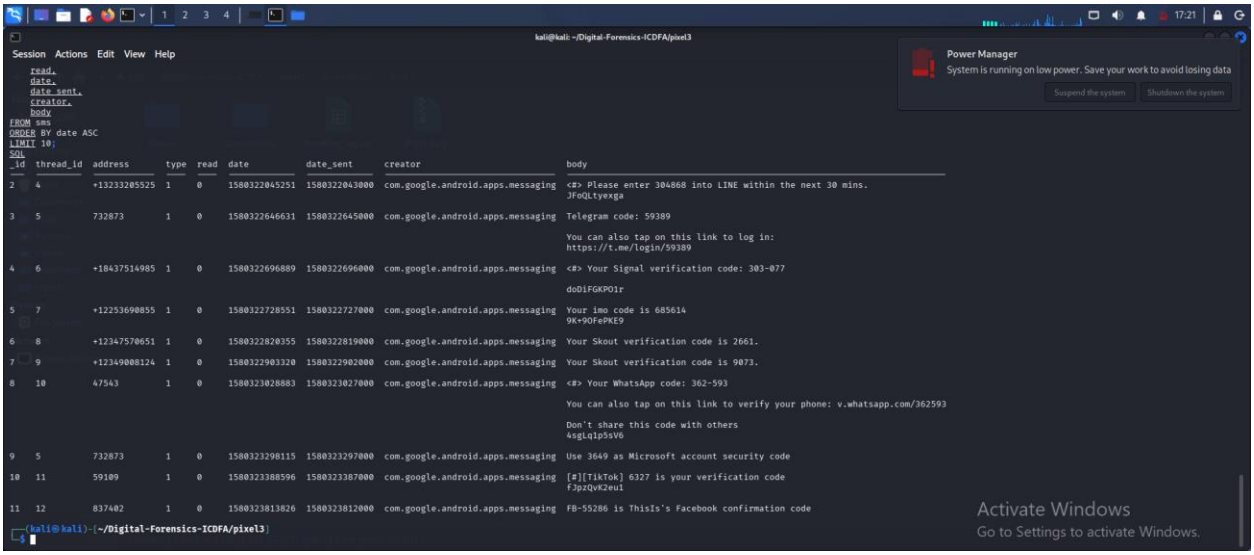
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$
```

```
d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827  ./Pixel 3/data/user_de/0/com.android.providers.telephony/databases/mmssms.db

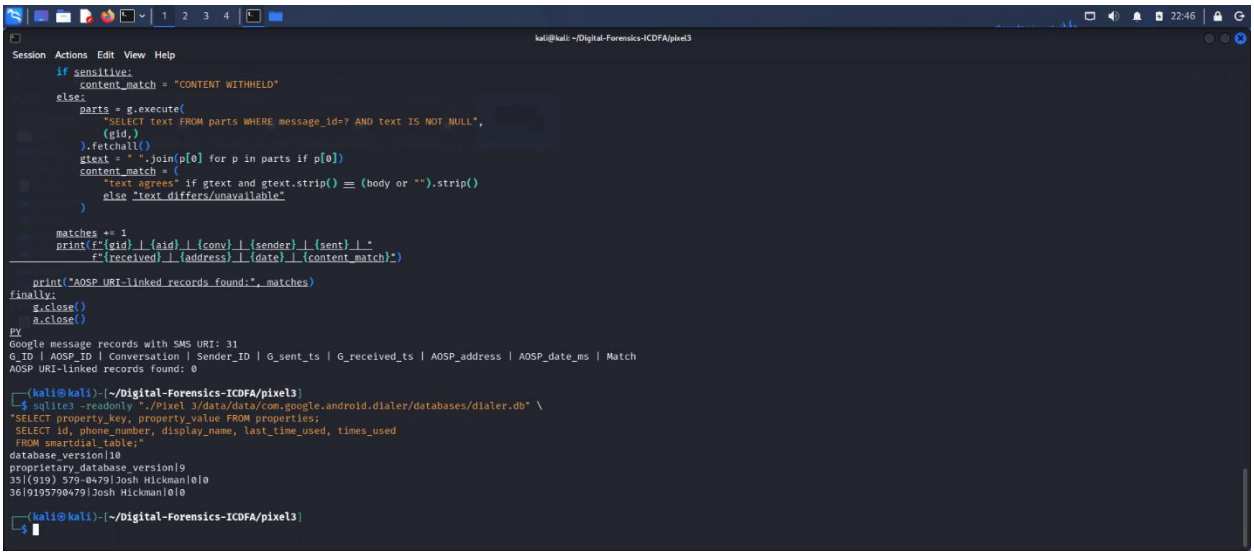
┌──(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
└─$ sqlite3 -readonly "./Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db" <<'SQL'
.tables
.schema sms
.schema pdu
.schema part
.schema threads
SELECT 'sms' AS table_name, COUNT(*) AS row_count FROM sms;
SELECT 'pdu' AS table_name, COUNT(*) AS row_count FROM pdu;
SELECT 'part' AS table_name, COUNT(*) AS row_count FROM part;
SELECT 'threads' AS table_name, COUNT(*) AS row_count FROM threads;
SQL
addr                pdu_restricted      threads
android_metadata    pending_msgs        words
attachments         rate                words_content
canonical_addresses raw                 words_segdir
drm                 sms                 words_segments
part                sms_restricted
pdu                 sr_pending
CREATE TABLE sms (_id INTEGER PRIMARY KEY,thread_id INTEGER,address TEXT,person INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,protocol INTEGER,read INTEGER DEFAULT 0,status INTEGER DEFAULT -1,type INTEGER,reply_path_present INTEGER,
subject TEXT,body TEXT,service_center TEXT,locked INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, error_code INTEGER DEFAULT 0,creator TEXT,seen INTEGER DEFAULT 0);
CREATE TRIGGER sms_update_thread_on_insert AFTER INSERT ON sms BEGIN  UPDATE threads SET    date = (strftime('%s','now') * 1000),     snippet = new.body,     snippet_cs = 0  WHERE threads._id = new.thread_id;   UPDATE threads SET messa
ge_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads      ON threads._id = thread_id      WHERE thread_id = new.thread_id       AND sms.type &  3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads      ON threads
._id = thread_id      WHERE thread_id = new.thread_id       AND (m_type=132 OR m_type=130 OR m_type=128)       AND msg_box &  3)  WHERE threads._id = new.thread_id;  UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM
 sms          WHERE read = 0          AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER sms_update_thread_date_subject_on_update AFTER  UPDATE OF date, body, type  ON sms BEGIN  UPDATE threads SET    date = (strftime('%s','now') * 1000),     snippet = new.body,     snippet_cs = 0  WHERE threads._id = new.th
read_id;  UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads      ON threads._id = thread_id      WHERE thread_id = new.thread_id       AND sms.type &  3) +      (SELECT COUNT(pdu._id) FROM pdu
 LEFT JOIN threads      ON threads._id = thread_id      WHERE thread_id = new.thread_id       AND (m_type=132 OR m_type=130 OR m_type=128)       AND msg_box &  3)  WHERE threads._id = new.thread_id;  UPDATE threads SET read =
CASE (SELECT COUNT(*)          FROM sms          WHERE read = 0          AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER sms_update_thread_read_on_update AFTER  UPDATE OF read  ON sms BEGIN  UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM sms          WHERE read = 0          AND thread_id = threads._id)      WHEN 0 THE
N 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER update_threads_error_on_update_sms  AFTER UPDATE OF type ON sms  WHEN (OLD.type &  5 AND NEW.type = 5)    OR (OLD.type = 5 AND NEW.type &  5) BEGIN  UPDATE threads SET error =     CASE      WHEN NEW.type = 5 THEN error
 + 1      ELSE error - 1    END  WHERE _id = NEW.thread_id; END;
CREATE TRIGGER sms_words_update AFTER UPDATE ON sms BEGIN UPDATE words  SET index_text = NEW.body WHERE (source_id=NEW._id AND table_to_use=1);  END;
CREATE TRIGGER sms_words_delete AFTER DELETE ON sms BEGIN DELETE FROM   words WHERE source_id = OLD._id AND table_to_use = 1; END;
CREATE INDEX typeThreadIdIndex ON sms (type, thread_id);
CREATE INDEX threadIdDateIndex ON sms (thread_id, date);
CREATE TABLE pdu (_id INTEGER PRIMARY KEY AUTOINCREMENT,thread_id INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,msg_box INTEGER,read INTEGER DEFAULT 0,m_id TEXT,sub TEXT,sub_cs INTEGER,ct_t TEXT,ct_l TEXT,exp INTEGER,m_cls TEXT,m_ty
pe INTEGER,v INTEGER,m_size INTEGER,pri INTEGER,rr INTEGER,rpt_a INTEGER,resp_st INTEGER,st INTEGER,tr_id TEXT,retr_st INTEGER,retr_txt TEXT,retr_txt_cs INTEGER,read_status INTEGER,ct_cls INTEGER,resp_txt TEXT,d_tm INTEGER,d_rpt INTEGE
R,locked INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, seen INTEGER DEFAULT 0,creator TEXT,text_only INTEGER DEFAULT 0);
CREATE TRIGGER part_cleanup DELETE ON pdu BEGIN   DELETE FROM part  WHERE mid=old._id;END;
CREATE TRIGGER addr_cleanup DELETE ON pdu BEGIN   DELETE FROM addr  WHERE msg_id=old._id;END;
CREATE TRIGGER cleanup_delivery_and_read_report AFTER DELETE ON pdu WHEN old.m_type=128 BEGIN   DELETE FROM pdu  WHERE (m_type=134    OR m_type=136)    AND m_id=old.m_id; END;
CREATE TRIGGER update_threads_on_update_pdu  AFTER UPDATE of thread_id ON pdu  BEGIN   UPDATE threads SET has_attachment=1 WHERE _id IN    (SELECT pdu.thread_id FROM part JOIN pdu      WHERE part.ct !='text/plain' AND part.ct != 'appl
```

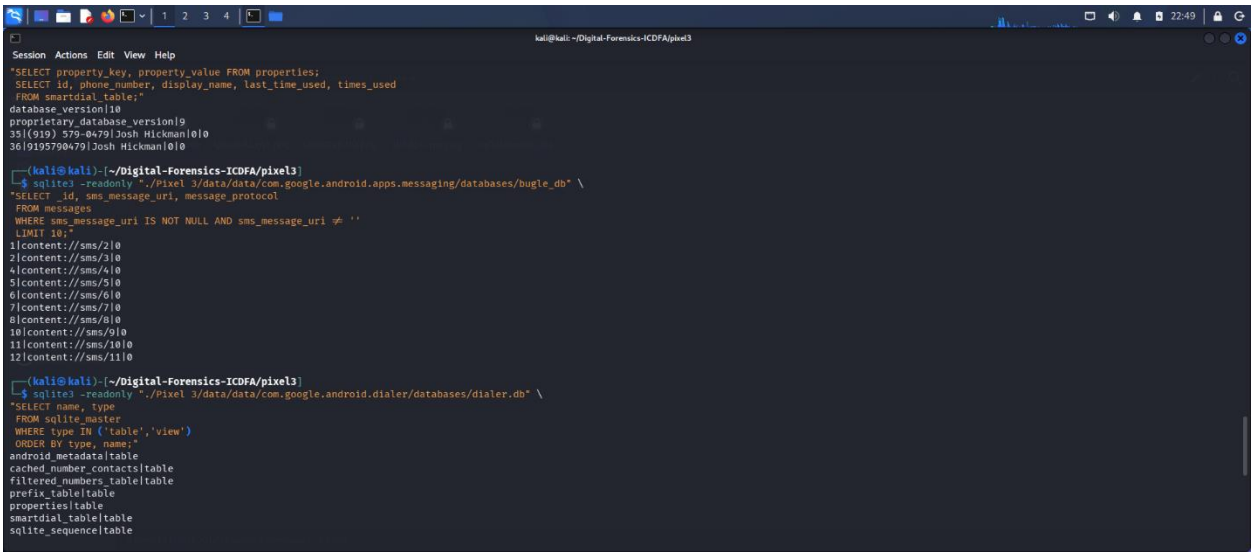


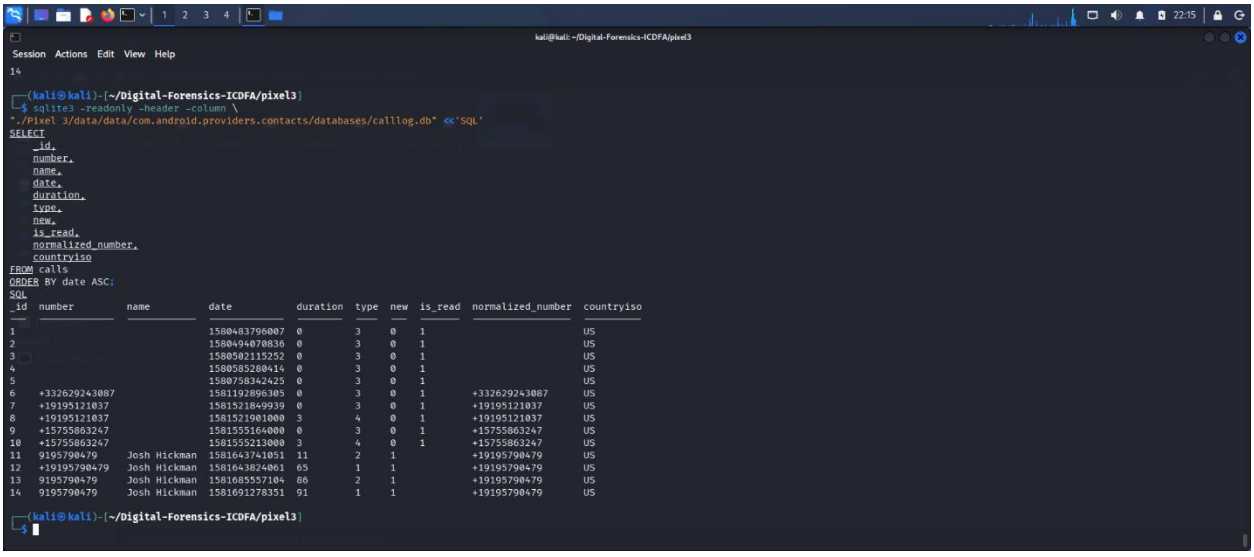


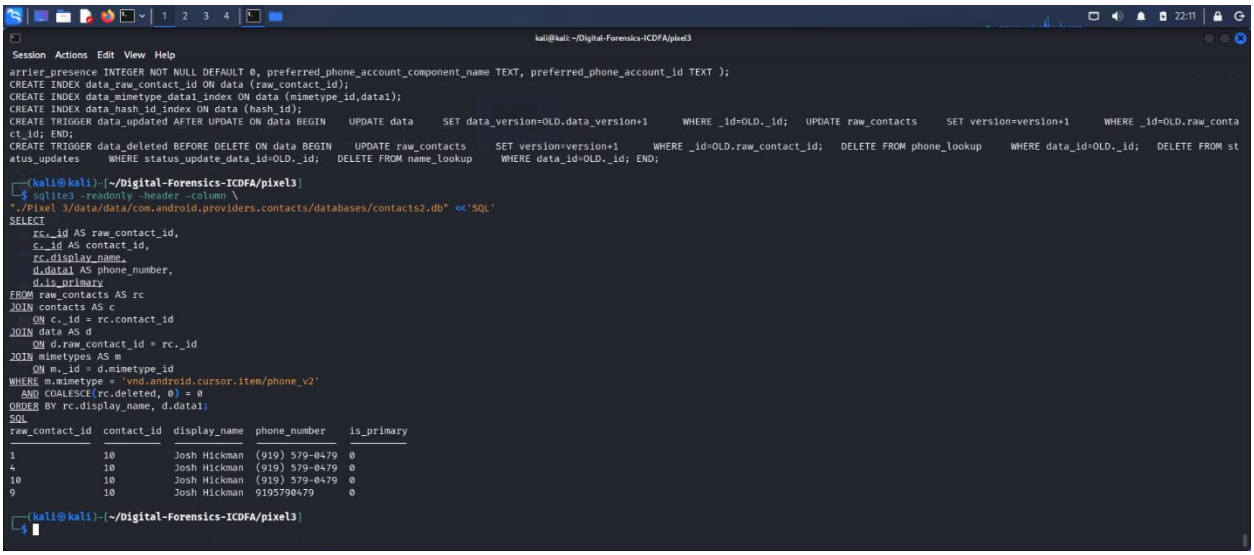


Task 5 Google Messages and Dialer

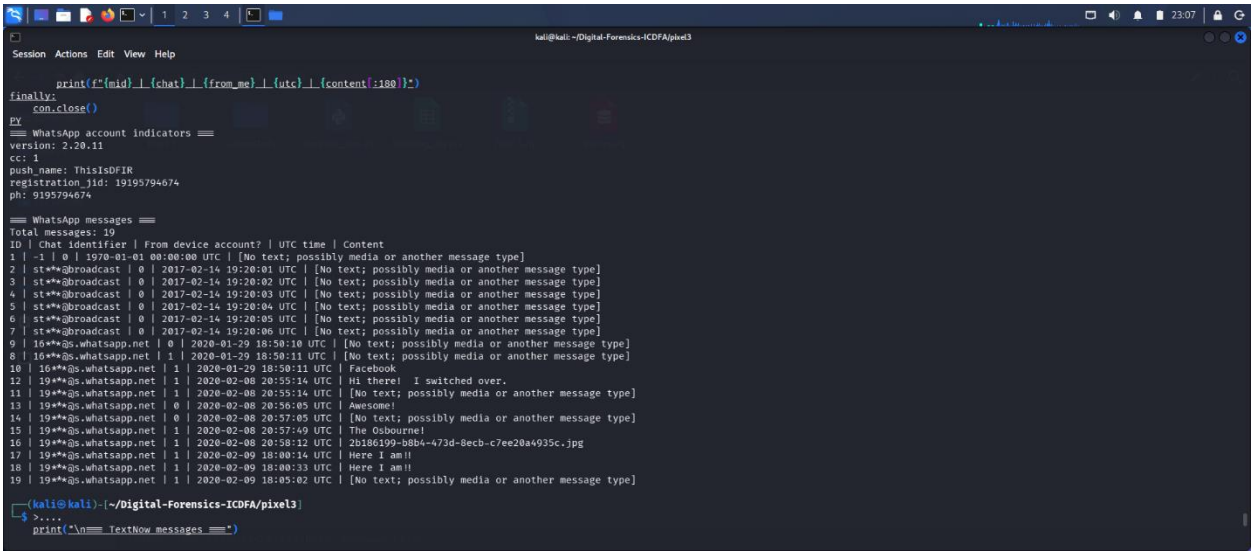

kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
if sensitive:
content_match = "CONTENT WITHHELD"
else:
parts = g.execute(
"SELECT text FROM parts WHERE message_id=? AND text IS NOT NULL",
(gid,)
).fetchall()
gtext = " ".join(p[0] for p in parts if p[0])
content_match = (
"text agrees" if gtext and gtext.strip() == (body or "").strip()
else "text differs/unavailable"
)
matches += 1
print(f"{gid} | {aid} | {conv} | {sender} | {sent} | "
f"{received} | {address} | {date} | {content_match}")
print("AOSP URI-linked records found:", matches)
finally:
g.close()
a.close()
PY
Google message records with SMS URI: 31
G_ID | AOSP_ID | Conversation | Sender_ID | G_sent_ts | G_received_ts | AOSP_address | AOSP_date_ms | Match
AOSP URI-linked records found: 0
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ sqlite3 -readonly "./Pixel 3/data/data/com.google.android.dialer/databases/dialer.db" \
"SELECT property_key, property_value FROM properties;
SELECT id, phone_number, display_name, last_time_used, times_used
FROM smartdial_table;"
database_version|10
proprietary_database_version|9
35|(919) 579-0479|Josh Hickman|0|0
36|9195790479|Josh Hickman|0|0
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$

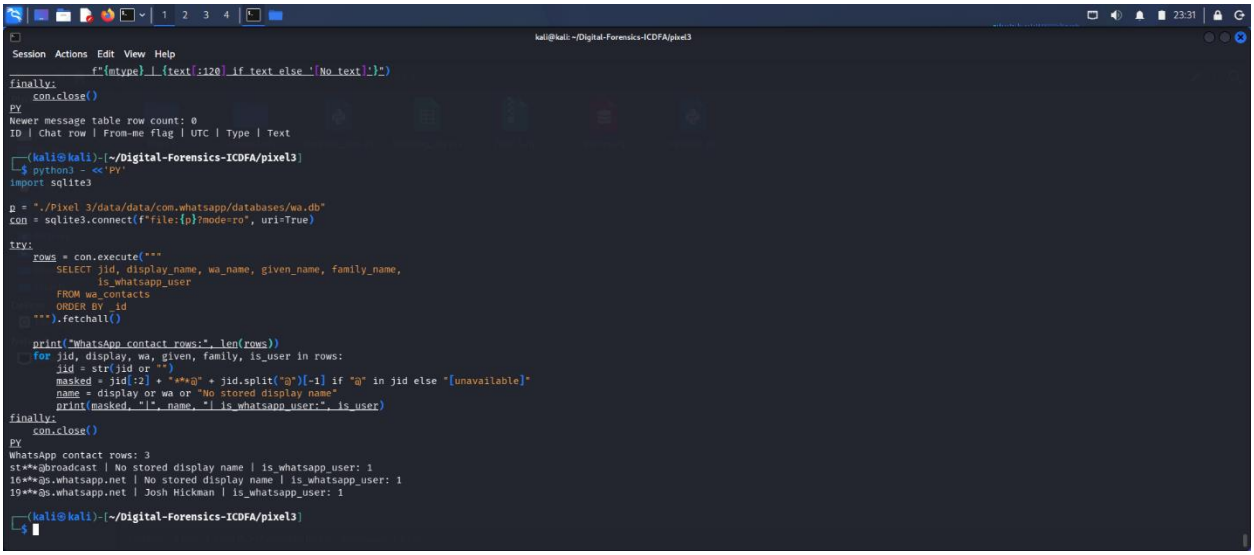

kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
"SELECT property_key, property_value FROM properties;
SELECT id, phone_number, display_name, last_time_used, times_used
FROM smartdial_table;"
database_version|10
proprietary_database_version|9
35|(919) 579-0479|Josh Hickman|0|0
36|9195790479|Josh Hickman|0|0
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ sqlite3 -readonly "./Pixel 3/data/data/com.google.android.apps.messaging/databases/bugle_db" \
"SELECT _id, sms_message_uri, message_protocol
FROM messages
WHERE sms_message_uri IS NOT NULL AND sms_message_uri != ''
LIMIT 10;"
1|content://sms/2|0
2|content://sms/3|0
4|content://sms/4|0
5|content://sms/5|0
6|content://sms/6|0
7|content://sms/7|0
8|content://sms/8|0
10|content://sms/9|0
11|content://sms/10|0
12|content://sms/11|0
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ sqlite3 -readonly "./Pixel 3/data/data/com.google.android.dialer/databases/dialer.db" \
"SELECT name, type
FROM sqlite_master
WHERE type IN ('table','view')
ORDER BY type, name;"
android_metadata|table
cached_number_contacts|table
filtered_numbers_table|table
prefix_table|table
properties|table
smartdial_table|table
sqlite_sequence|table


kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
14
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ sqlite3 -readonly -header -column \
"./Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db" <<'SQL'
SELECT
_id,
number,
name,
date,
duration,
type,
new,
is_read,
normalized_number,
countryiso
FROM calls
ORDER BY date ASC;
SQL
_id number name date duration type new is_read normalized_number countryiso
1 1580483796007 0 3 0 1 US
2 1580494070836 0 3 0 1 US
3 1580502115252 0 3 0 1 US
4 1580585280414 0 3 0 1 US
5 1580758342425 0 3 0 1 US
6 +332629243087 1581192896305 0 3 0 1 +332629243087 US
7 +19195121037 1581521849939 0 3 0 1 +19195121037 US
8 +19195121037 1581521901000 3 4 0 1 +19195121037 US
9 +15755863247 1581555164000 0 3 0 1 +15755863247 US
10 +15755863247 1581555213000 3 4 0 1 +15755863247 US
11 9195790479 Josh Hickman 1581643741051 11 2 1 +19195790479 US
12 +19195790479 Josh Hickman 1581643824061 65 1 1 +19195790479 US
13 9195790479 Josh Hickman 1581685557104 80 2 1 +19195790479 US
14 9195790479 Josh Hickman 1581691278351 91 1 1 +19195790479 US
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$

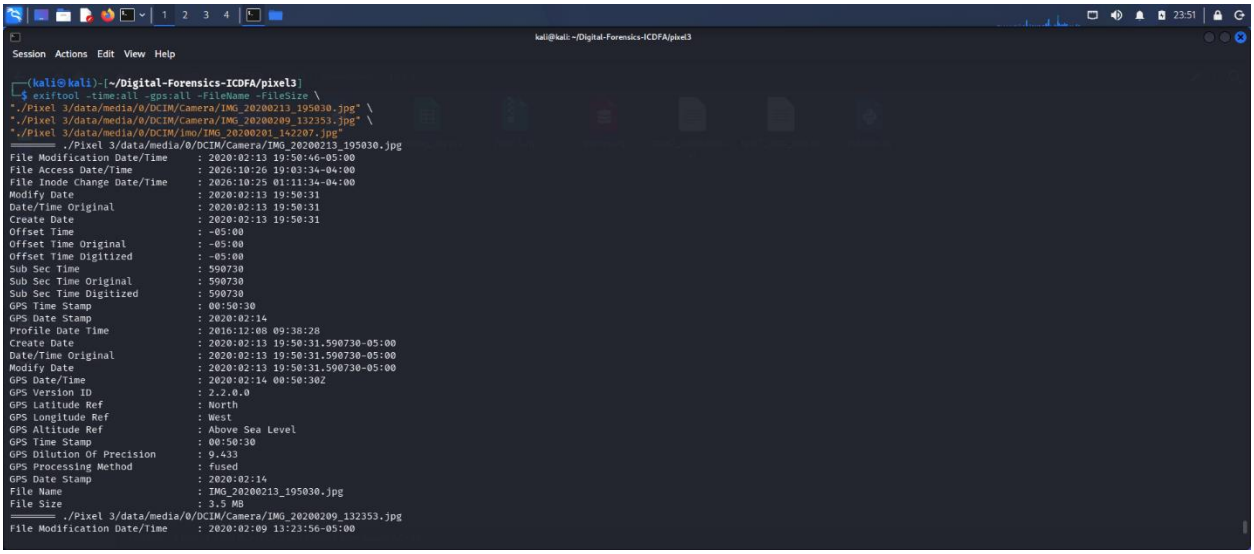


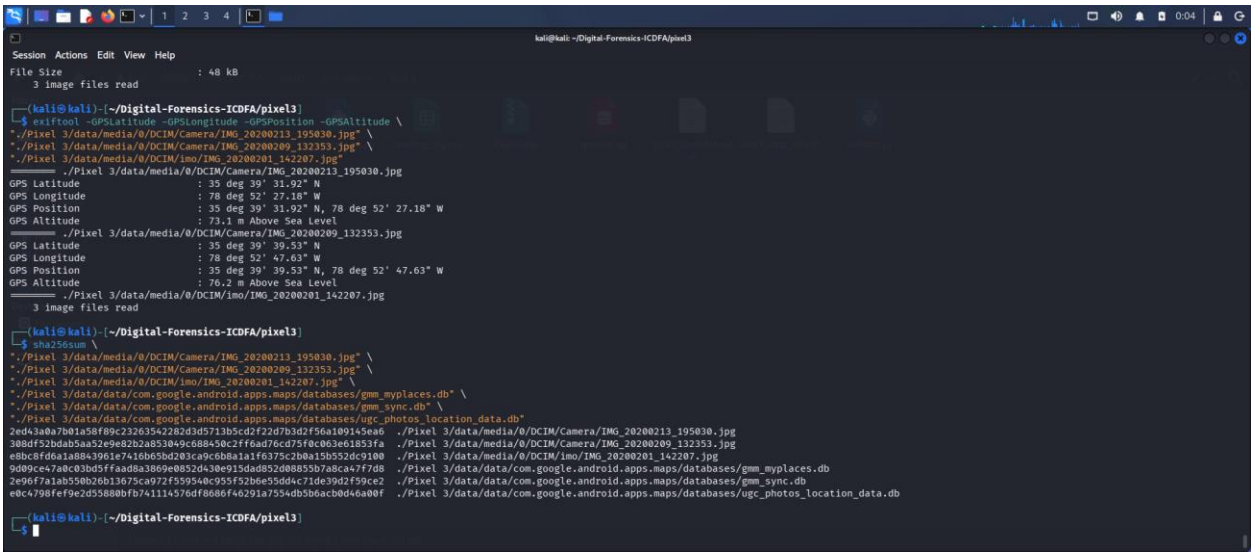
Task 6 WhatsApp and another application





Task 7 Maps and photographs


kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ exiftool -time:all -gps:all -FileName -FileSize \
"./Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg" \
"./Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg" \
"./Pixel 3/data/media/0/DCIM/imo/IMG_20200201_142207.jpg"
======== ./Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg
File Modification Date/Time : 2020:02:13 19:50:46-05:00
File Access Date/Time : 2026:10:26 19:03:34-04:00
File Inode Change Date/Time : 2026:10:25 01:11:34-04:00
Modify Date : 2020:02:13 19:50:31
Date/Time Original : 2020:02:13 19:50:31
Create Date : 2020:02:13 19:50:31
Offset Time : -05:00
Offset Time Original : -05:00
Offset Time Digitized : -05:00
Sub Sec Time : 590730
Sub Sec Time Original : 590730
Sub Sec Time Digitized : 590730
GPS Time Stamp : 00:50:30
GPS Date Stamp : 2020:02:14
Profile Date Time : 2016:12:08 09:38:28
Create Date : 2020:02:13 19:50:31.590730-05:00
Date/Time Original : 2020:02:13 19:50:31.590730-05:00
Modify Date : 2020:02:13 19:50:31.590730-05:00
GPS Date/Time : 2020:02:14 00:50:30Z
GPS Version ID : 2.2.0.0
GPS Latitude Ref : North
GPS Longitude Ref : West
GPS Altitude Ref : Above Sea Level
GPS Time Stamp : 00:50:30
GPS Dilution Of Precision : 9.433
GPS Processing Method : fused
GPS Date Stamp : 2020:02:14
File Name : IMG_20200213_195030.jpg
File Size : 3.5 MB
======== ./Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg
File Modification Date/Time : 2020:02:09 13:23:56-05:00


kali@kali: ~/Digital-Forensics-ICDFA/pixel3
Session Actions Edit View Help
File Size : 48 kB
3 image files read
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ exiftool -GPSLatitude -GPSLongitude -GPSPosition -GPSAltitude \
"./Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg" \
"./Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg" \
"./Pixel 3/data/media/0/DCIM/imo/IMG_20200201_142207.jpg"
======== ./Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg
GPS Latitude : 35 deg 39' 31.92" N
GPS Longitude : 78 deg 52' 27.18" W
GPS Position : 35 deg 39' 31.92" N, 78 deg 52' 27.18" W
GPS Altitude : 73.1 m Above Sea Level
======== ./Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg
GPS Latitude : 35 deg 39' 39.53" N
GPS Longitude : 78 deg 52' 47.63" W
GPS Position : 35 deg 39' 39.53" N, 78 deg 52' 47.63" W
GPS Altitude : 76.2 m Above Sea Level
======== ./Pixel 3/data/media/0/DCIM/imo/IMG_20200201_142207.jpg
3 image files read
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
$ sha256sum \
"./Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg" \
"./Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg" \
"./Pixel 3/data/media/0/DCIM/imo/IMG_20200201_142207.jpg" \
"./Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_myplaces.db" \
"./Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_sync.db" \
"./Pixel 3/data/data/com.google.android.apps.maps/databases/ugc_photos_location_data.db"
2ed43a0a7b01a58f89c2326354228 2d3d5713b5cd2f22d7b3d2f56a109145ea6 ./Pixel 3/data/media/0/DCIM/Camera/IMG_20200213_195030.jpg
308df52bdab5aa52e9e82b2a853049c688450c2ff6ad76cd75f0c063e61853fa ./Pixel 3/data/media/0/DCIM/Camera/IMG_20200209_132353.jpg
e8bc8fd6a1a8843961e7416b65bd203ca9c6b8a1af6375c2b0a15b552dc9100 ./Pixel 3/data/media/0/DCIM/imo/IMG_20200201_142207.jpg
9d09ce47a0c03bd5ffaad8a3869e0852d430e915da852d08855b7a8ca47f7d8 ./Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_myplaces.db
2e96f7a1ab550b26b13675ca972f559540c955f52b6e55dd4c71de39d2f59ce2 ./Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_sync.db
e0c4798fef9e2d55880bfb741114576df8686f462911a7554db5b6acb0d46a00f ./Pixel 3/data/data/com.google.android.apps.maps/databases/ugc_photos_location_data.db
(kali㉿kali)-[~/Digital-Forensics-ICDFA/pixel3]
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